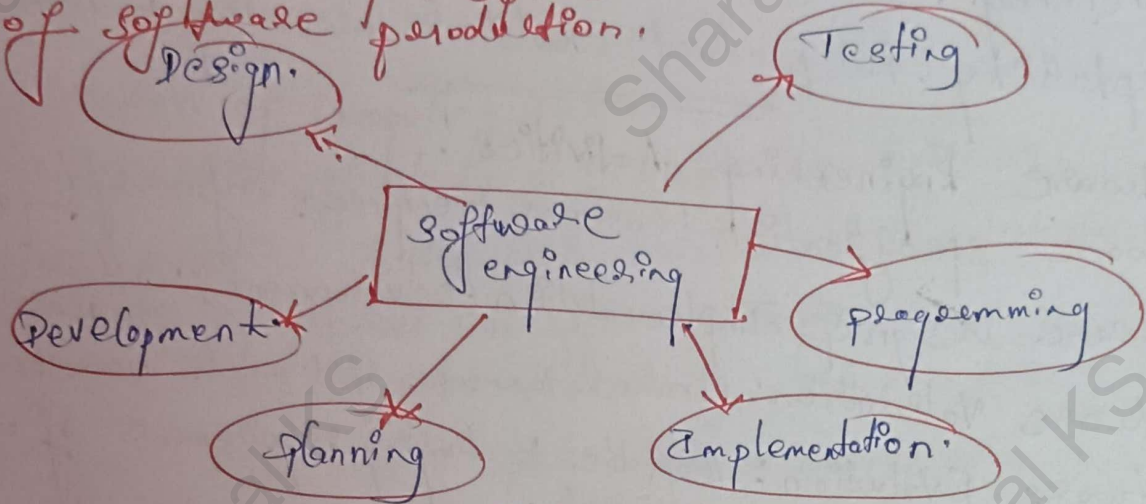


① INTRODUCTION.

Software Engineering: Developing software to solve a problem which is cost-effective.

→ It is concerned with theories, methods & tools for professional software development.

→ It is an engineering discipline that is concerned with all aspects of software production.



Importance of Software Engineering:

- ① Decrease time.
- ② Reduced complexity.
- ③ minimize software cost
- ④ Handling Big project.
- ⑤ Effectiveness
- ⑥ Reliable software.

Project management: It is the process of planning & implementing a project to meet certain objectives by following a pre-determined schedule & action plan.

Difference between Generic & Customized products.

Generic Product	Customized product:
① Stand-alone system that are marketed & sold to any customer who wishes to buy.	① Software that is commissioned by a specific customer to meet their own needs.
② The specifications of what software does & software changes done are made by developer.	② Specifications of what software does & software changes are done by customer of software.
③ Ex: CAD software, graphic programs etc.	③ Ex: Embedded control system, traffic control software etc.

* **Software**: It is a computer programs & associated documentation.

* **Essential attributes of a Good Software**: (fundamentals)

- ① maintainability: meeting expectations of users.
- ② Dependability: cost, reliability & safety/security.
- ③ Efficiency: How well the software performs.
- ④ Acceptability: End product should be according to user need.

* **Software Engineering Activities**:

- ① Software specification: Customer & engineers define software requirements.
- ② Software Design & Implementation (Development): Software designed & programmed.
- ③ Software Validation: Checked to ensure if it meets user requirements.
- ④ Software Evolution: Adaptable to changing requirements of users & market.

* **Difference between**:

① **Software Engineering & Computer Science**:

- Computer science focus on theory & fundamentals
- Software Engineering focus on practicalities of developing & ~~not~~ delivering useful software.

② **Software Engineering & System Engineering**:

- System engineering is concerned with all aspects of computer-based systems development including hardware, software & process engineering.
- Software engineering is part of this more general process.

* **Key challenges Facing SE**: (Ethical Responsibilities)

- ① Managing complexity
- ② Changing Requirements
- ③ Security & privacy
- ④ Ensuring quality & reliability
- ⑤ meeting deadlines
- ⑥ Resource management
- ⑦ Integration with existing system
- ⑧ Scalability
- ⑨ keeping up with rapid technology changes.
- ⑩ Cost control
- ⑪ Testing & validation
- ⑫ Communication & collaboration.
- ⑬ Ethical & Social consideration
- ⑭ maintenance & upgrade
- ⑮ User Experience

* **Cost of SE**: Roughly 60% : Development cost.

40% : Testing cost.

→ For custom software evolution costs often exceeds development cost.

* **Engineering Discipline**: using appropriate theories & methods to solve problems bearing in mind organizational & financial constraints.

* **ACM/IEEE CODE of ETHICS**:

→ Association of Computing Machinery.

→ Institute of Electrical & Electronics Engineers.

→ This code contains 8 principles related to the behaviour of & decisions made by professional software engineers.

- ① **PUBLIC**: SE's shall act consistently with the public interest.
- ② **CLIENT & EMPLOYER**: SE's shall act in a manner that is in the best interests of their client & employer consistent with public interest.
- ③ **PRODUCT**: SE's shall ensure that their products & related modifications meet the highest professional standards possible.
- ④ **JUDGEMENT**: SE's shall maintain integrity & independence in their professional judgement.
- ⑤ **MANAGEMENT**: SE's managers & leaders shall subscribe to & promote an ethical approach to the management of software development & maintenance.
- ⑥ **PROFESSION**: SE's shall advance the integrity & reputation of the profession consistent with public interest.
- ⑦ **COLLEAGUES**: SE's shall be fair to & supportive of their colleagues.
- ⑧ **SELF**: SE's shall participate in lifelong learning regarding the practice of their profession & shall promote an ethical approach to the practice of the profession.

CASE STUDY

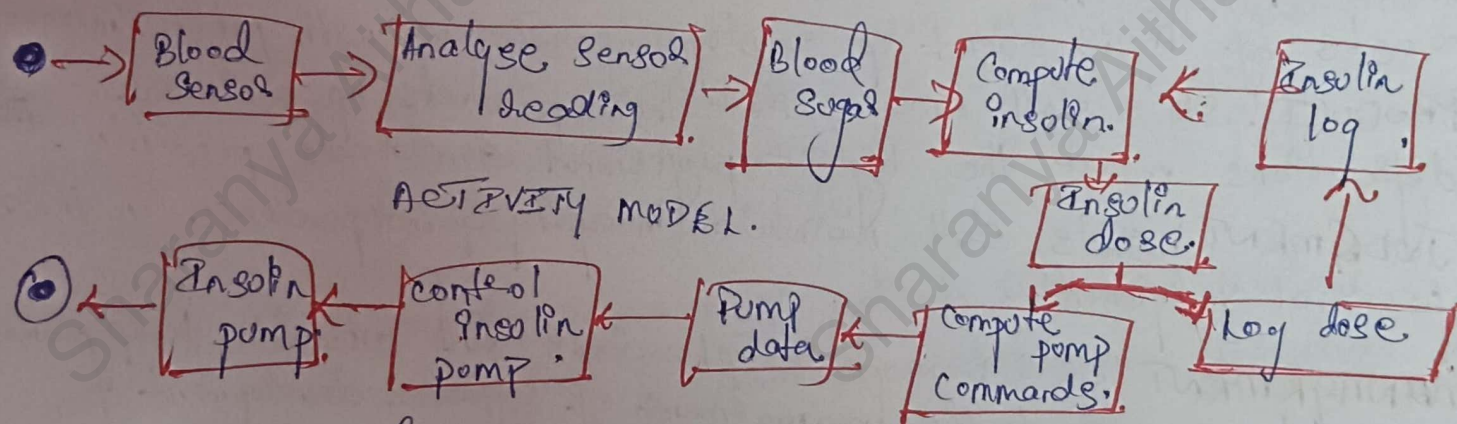
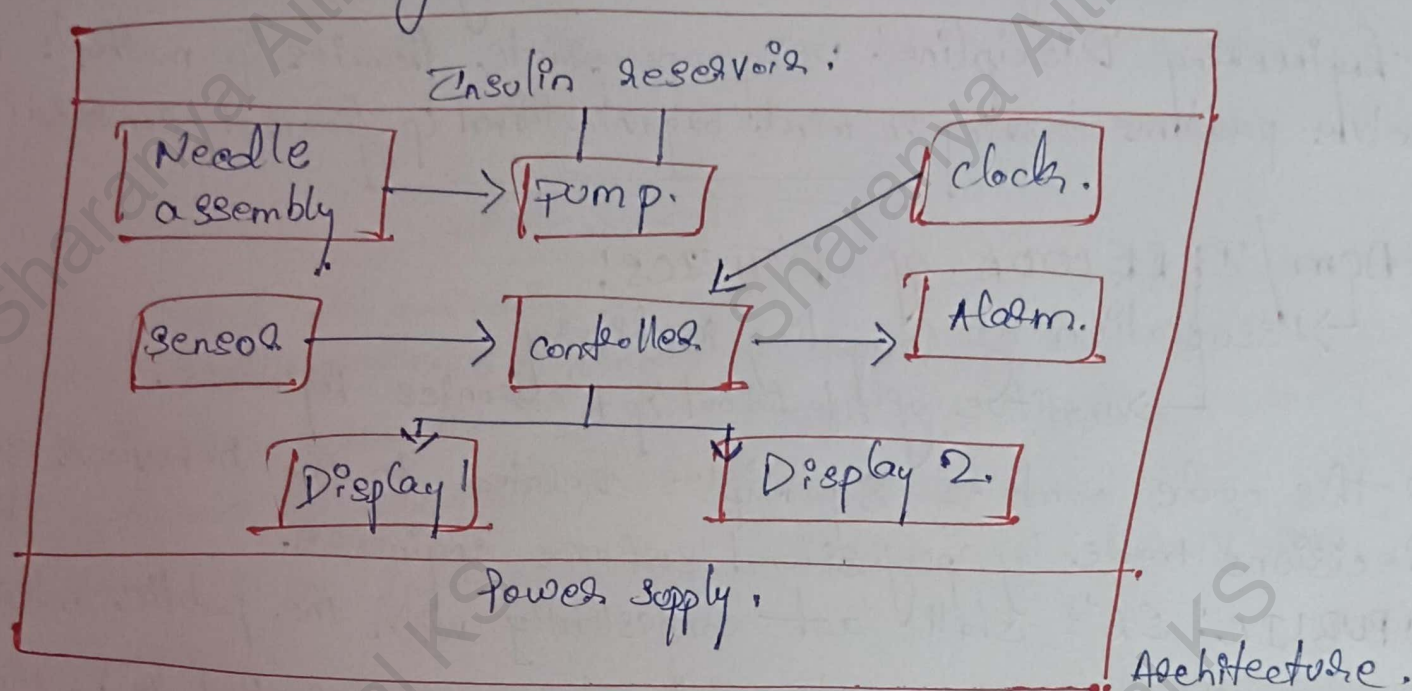
① **Angolin Pump**

② **MHCms**

③ **Weather station**

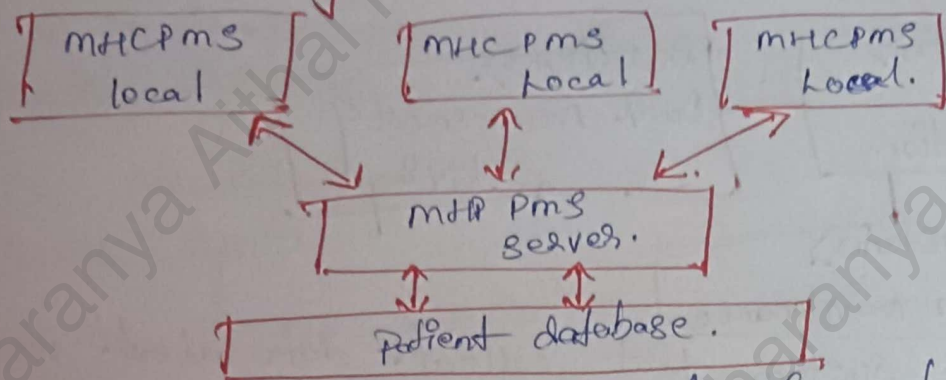
① PERSONAL INSULIN PUMP:

→ An embedded system in an insulin pump used by diabetics to maintain blood glucose control.



- collects data from a blood sugar sensor & calculates the amount of insulin required to be injected.
- Calculation based on rate of change of blood sugar level.
- sends signal to micro-pump to deliver the correct dose of insulin.
- It is a safety critical system.
- System shall be available to deliver insulin when required.
- It shall perform reliably & deliver the correct amount of insulin to counteract the current level of blood sugar.
- System is designed & implemented to ensure that all requirements are met always.

② MHC-PMS: organization!



A system used to maintain records of people receiving care for mental health problems & treatment they have received.

→ This maintains information about patients suffering from mental health problems & treatments that they have received.

→ MHC-PMS is an information system that is used in clinics.

→ Key features:

① Individual case management!

Clinicians can create records for patients, edit the information in the system, view patient history etc. The system supports data summary so that doctors can quickly learn about the key problems & treatments that have been prescribed.

② Patient monitoring!

This system monitors the records of patients that are involved in treatment & issues warnings if possible problems are detected.

③ Administrative reporting!

The system generates monthly management reports showing the number of patients treated at each clinic, patients entered & left the care system, the drugs prescribed & their cost etc.

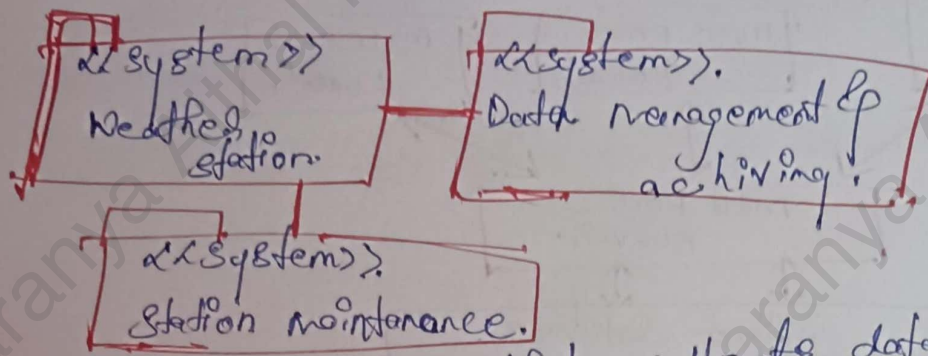
④ Privacy!

It is essential that patient information is confidential & is never disclosed to anyone apart from authorised medical staff & the patient themselves.

⑤ Safety!

The system should warn medical staff about potentially suicidal or dangerous patients. It should be available when need or safety might be compromised & it may be impossible to prescribe correct medication to patients.

③ Weather station: station's Environment:



A data collection system that collects data about weather conditions in remote areas.

→ It measures temperature, pressure, sunshine, rainfall, wind speed & wind direction.

① Weather station system:

Responsible for collecting weather data, carrying out some initial data processing & transmitting it to the data management system.

② Data management & archiving system:

It collects the data from all of the weather stations, carries out data processing & analysing & archives the data.

③ Station maintenance system:

This communicates by satellite with all weather stations to monitor the health of these systems & provide report of problems.

→ They support dynamic reconfiguration.

→ They manage system power, ensuring that batteries are charged & also shut down when there is potentially damaging weather.

~~~~~



## ② SOFTWARE PROCESSES.

It is a roadmap that provides a 'framework' for building high quality software.

→ It is structured set of activities required to develop a software system.

### \* Software process Activities:

① Specification: Define what system should do.

② Design & Implementation: organisation & implementation of system.

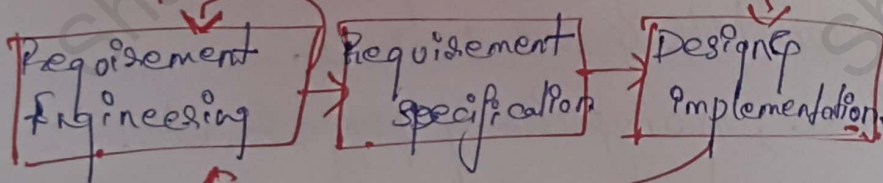
③ Validation: checking that it does what customers want.

④ Evolution: changing the system according to customer needs.

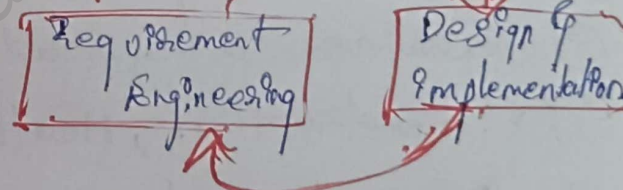
\* Plan driven process: These are the processes where all of process activities are planned in advance & progress is measured against this plan.

\* Agile process: These are incremental & it is easier to change the process to reflect changing customer requirements.

#### Plan driven:



#### Agile process:



#### Plan driven

→ Deals better with quality or nonfunctional requirements such as reliability & scalability.

→ All requirement are well defined at first stage

→ User interaction is less

→ Slower software development.

→ User gets product at the last stage

#### Agile process.

→ Not so good in quality or nonfunctional requirement compared to plan driven.

→ Not all requirement are necessarily defined.

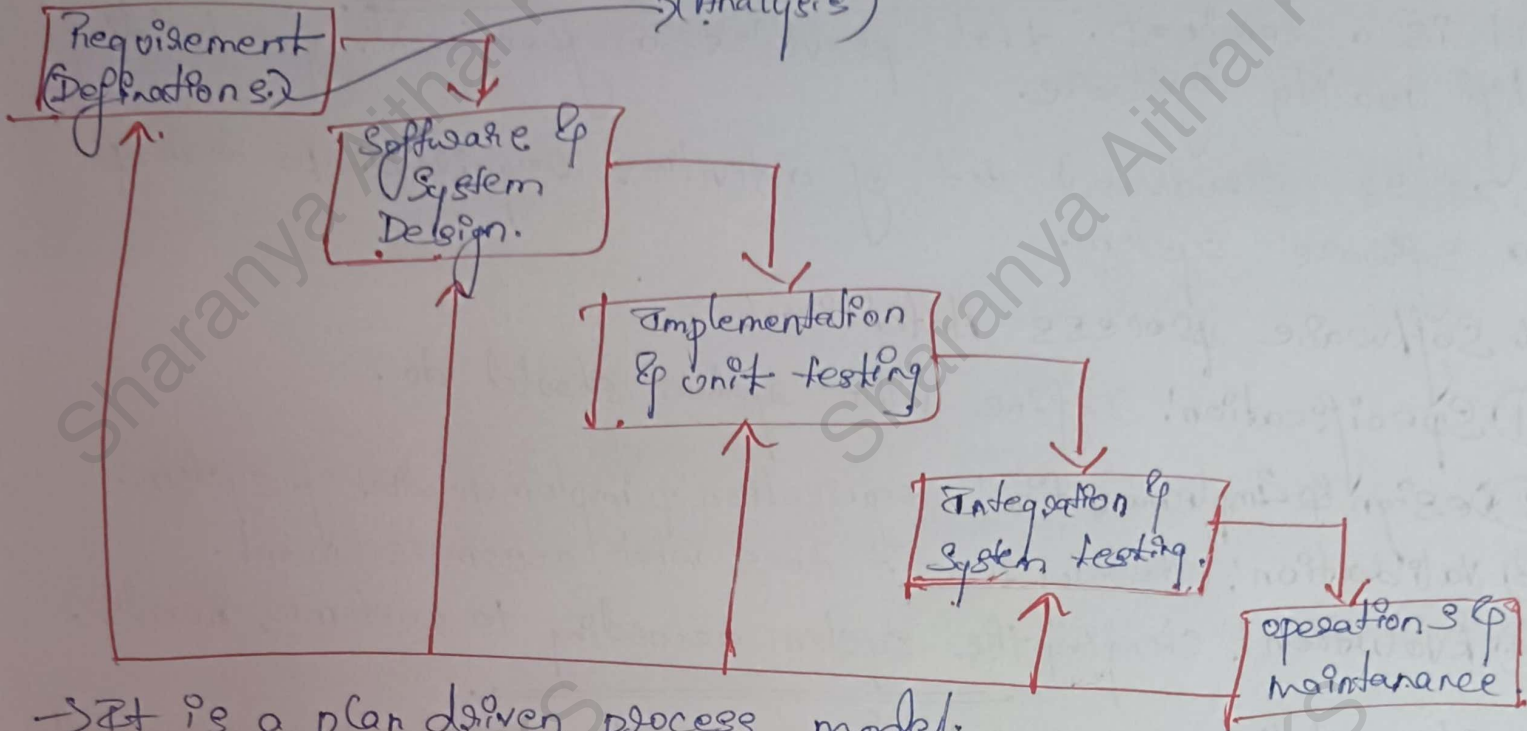
→ User interaction is more.

→ Rapid software development.

→ User gets the product at 1st version.



## WATER FALL MODEL:



→ It is a plan driven process model.

⇒ When to use waterfall model!:

- ① Application you need to develop is small & simple.
- ② Specifications are concise & do not change over time.
- ③ Requirements are concise & easy to understand.
- ④ Tools and techniques used are stable & not dynamic.
- ⑤ Resources are readily available & well trained.

⇒ Ex: Construction, Healthcare, manufacturing projects etc..

⇒ Advantages:

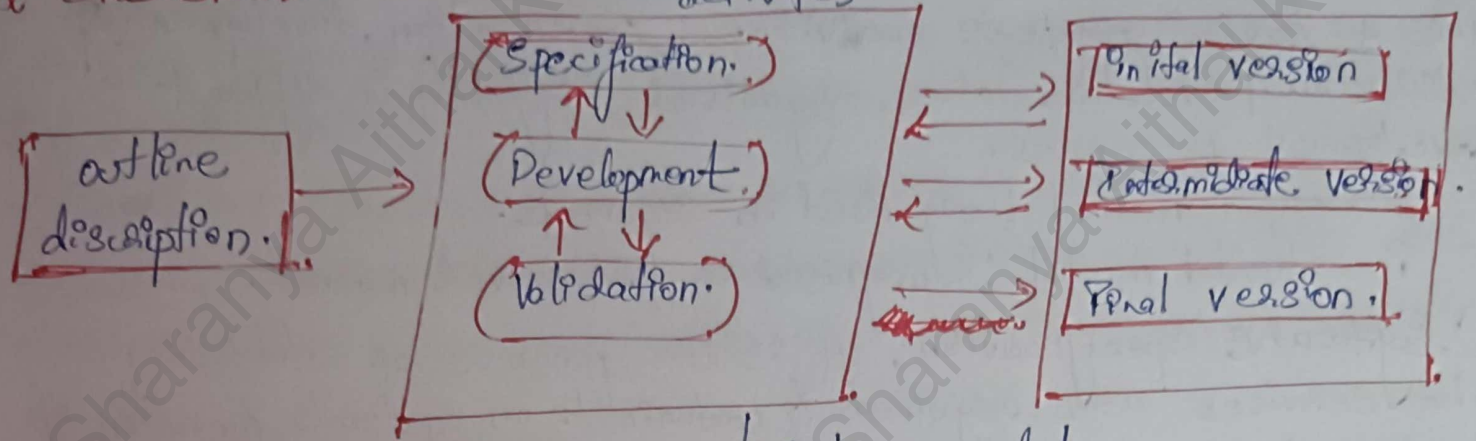
- ① Simple & easy to understand.
- ② Concise & well defined.
- ③ Clear milestones.
- ④ Well documented.
- ⑤ Estimate products final cost.
- ⑥ Estimate products release date.

⇒ Disadvantages:

- ① No mechanism for error correction.
- ② Do not accept changes in requirements.
- ③ No overlapping phases.



## Incremental Models: concurrent activities.



- This may be plan driven / Agile model.
- System development is broken down into many mini development projects.
- partial systems are successively built to produce a final total system.
- Highest priority system is tackled (completed) first.
- Product is developed incrementally.

### ⇒ Benefits:

- ① cost of accommodating changing customer requirement is reduced.
- ② Easier to get customer feedback on the development work that has been done.
- ③ more rapid delivery & deployment of useful software to the customer is possible.

### ⇒ Problems:

- ① process is not visible.
- ② System structure tends to degrade as new increments are added.

### ⇒ Advantages:

- ① software generated quickly.
- ② flexible & less expensive.
- ③ changes can be done throughout process.
- ④ less costly compared to other model.
- ⑤ Customer can give feedback.
- ⑥ Error can be easily identified.

### ⇒ Disadvantages:

- ① Resource intensive & incomplete product at each stage.
- ② Difficult to manage changes in large project & not suitable for small project.
- ③ Frequent client involvement needed.
- ④ more complex to manage.
- ⑤ High dependency on skilled team members.
- ⑥ Testing & debugging multiple times is time consuming.



## \* Process Activities:

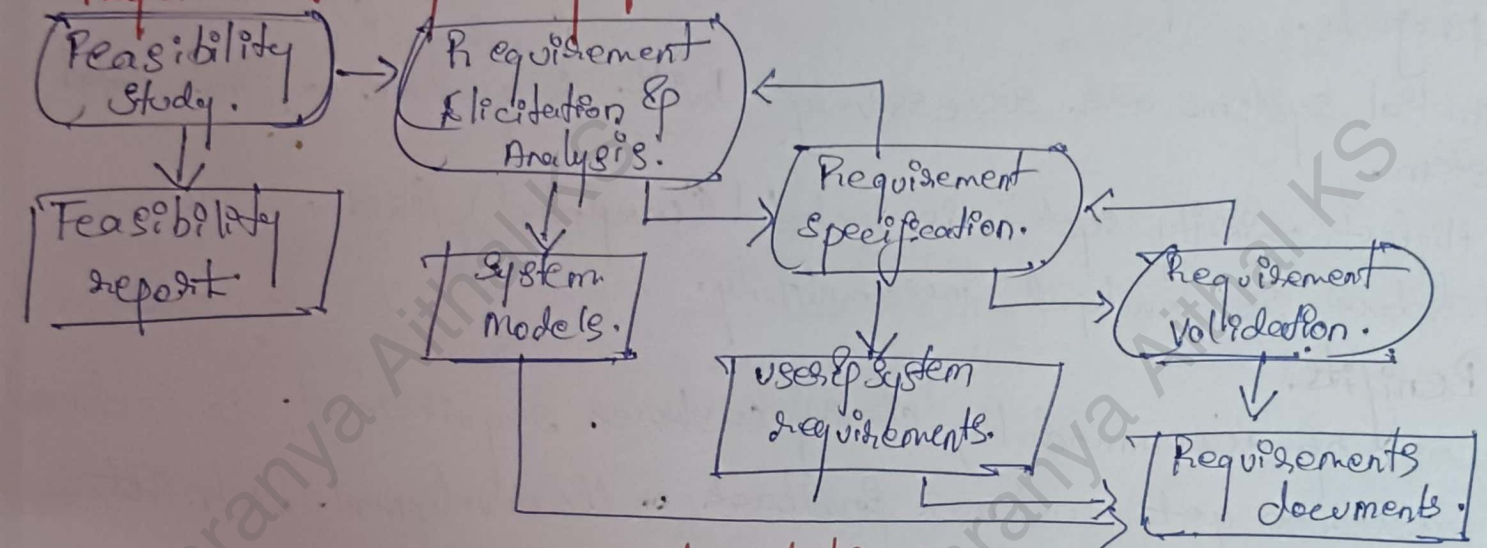
The 4 basic process activities: specification, development, & validation & evolution are organised differently in different development processes.

Waterfall model: organised in sequence.

Incremental model: organised in interleaved manner.

① **Software Specification:** It is the process of establishing what services are required & constraints on system's operation & development.

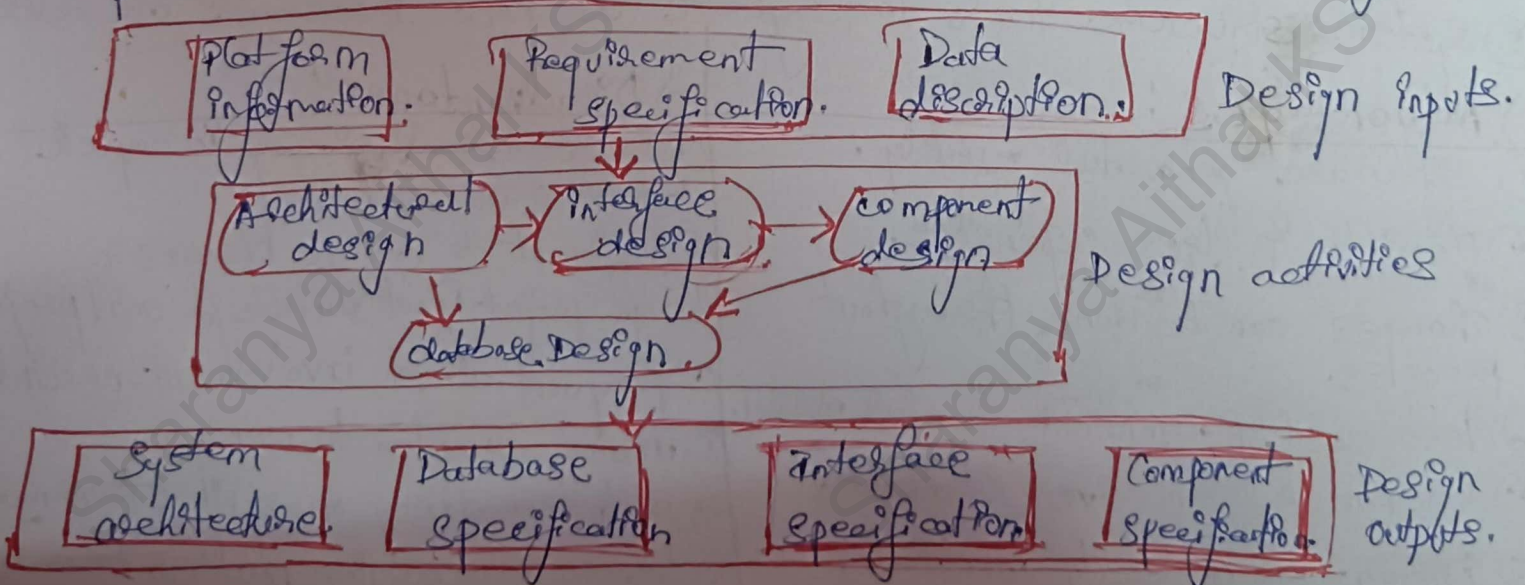
⇒ **Requirement Engineering Process:**



② **Software design & implementation:** It is the process of converting system specification into an executable system.

→ **Software Design:** Design a software structure that realises the specification.

→ **Implementation:** Translate this structure into executable program.

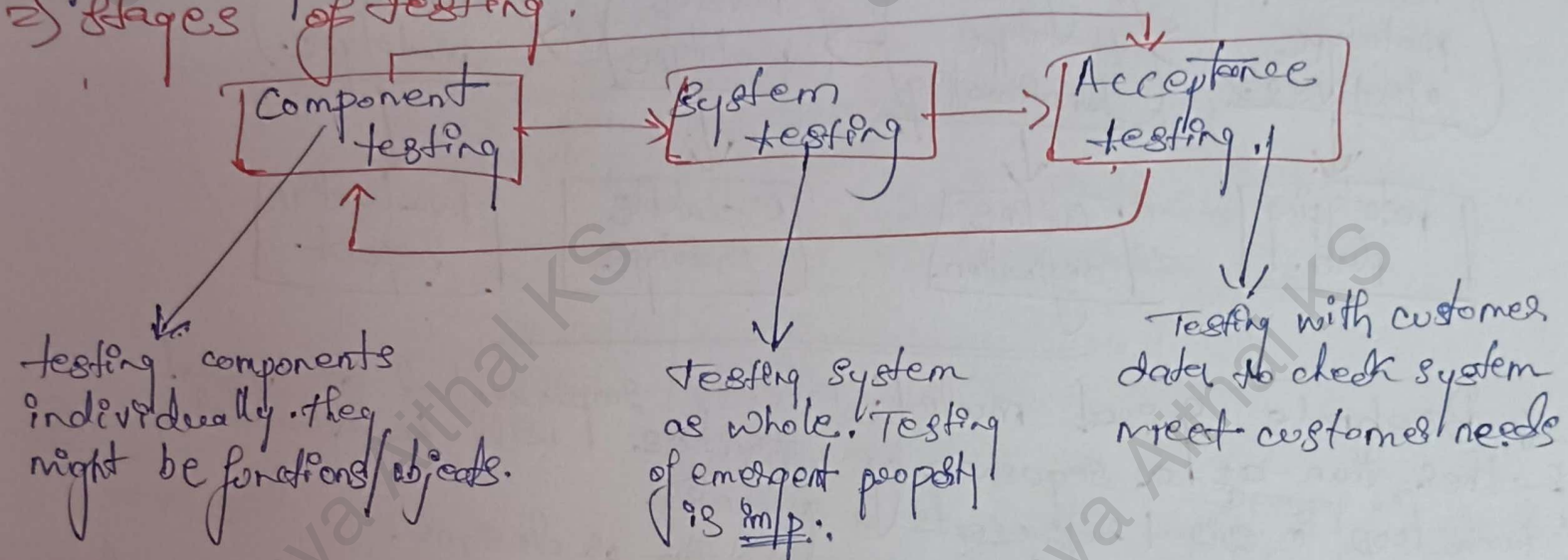




③ **Software Validation**: Verification & Validation is intended to show that a system conforms to its specification & meets the requirements of system customer.

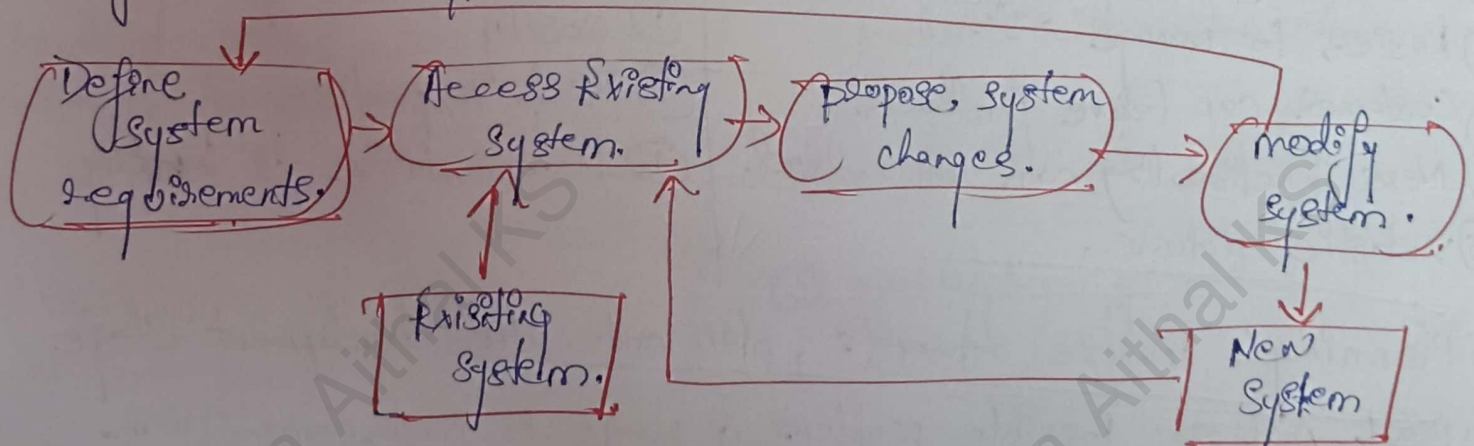
- Involves checking & review processes & system testing.
- Testing involves executing system with test cases that are derived from the specification of real data to be processed by the system.

④ **Stages of testing**:



④ **Software Evolution**:

- Software is inherently flexible & can change.
- As requirements change through changing business circumstances, the software that supports business must also evolve & change.

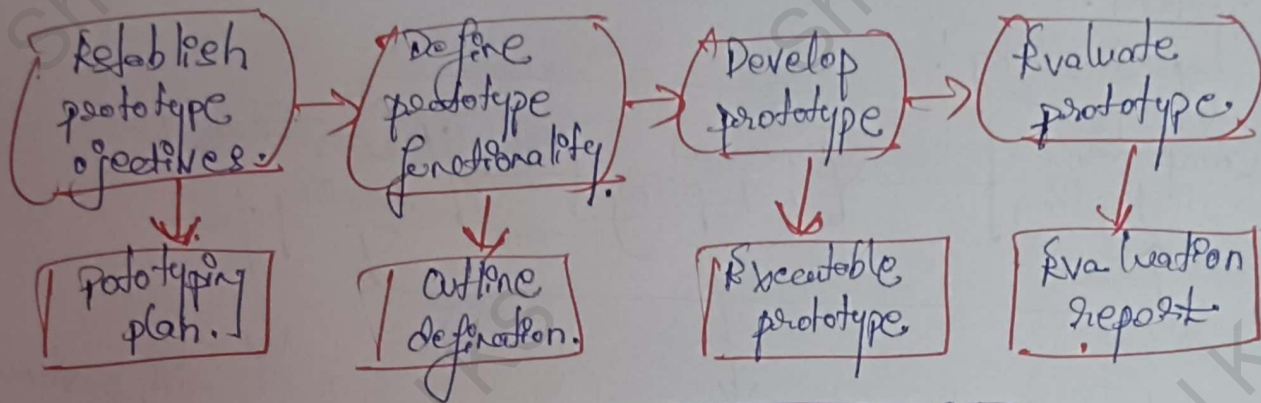


\* **Software prototyping**: A prototype is initial version of the system used to demonstrate concepts & try out design options.



## ⇒ Benefits:

- ① Improved system usability.
- ② A closer match to user real needs.
- ③ Improved design quality.
- ④ Improved maintainability.
- ⑤ Reduced development process.



**X Boehm's Spiral model:** Process represented as spiral rather than as a sequence of activities. with backstepping.

- Each loop in spiral represents a process.
- No fixed phases such as specification or design.
- Use: large project, creating prototype, evaluate risk & cost of projects.

## ⇒ Advantages:

- ① Better cost estimation.
- ② Easier to manage risk.
- ③ Customer can leave feedback.
- ④ New functionality can be added.
- ⑤ Relastic picture.

## ⇒ Disadvantages:

- ① Time consuming
- ② Costly
- ③ Rich Expense
- ④ Documentation is lengthy.

- ① **Planning:** Analyze demands, plan out the development stage.
- ② **Risk Analysis:** Possible problems & ways to overcome them.
- ③ **Development:** start working on the project.
- ④ **Evaluation:** When stage finished, client access results & provides the team with feedback.

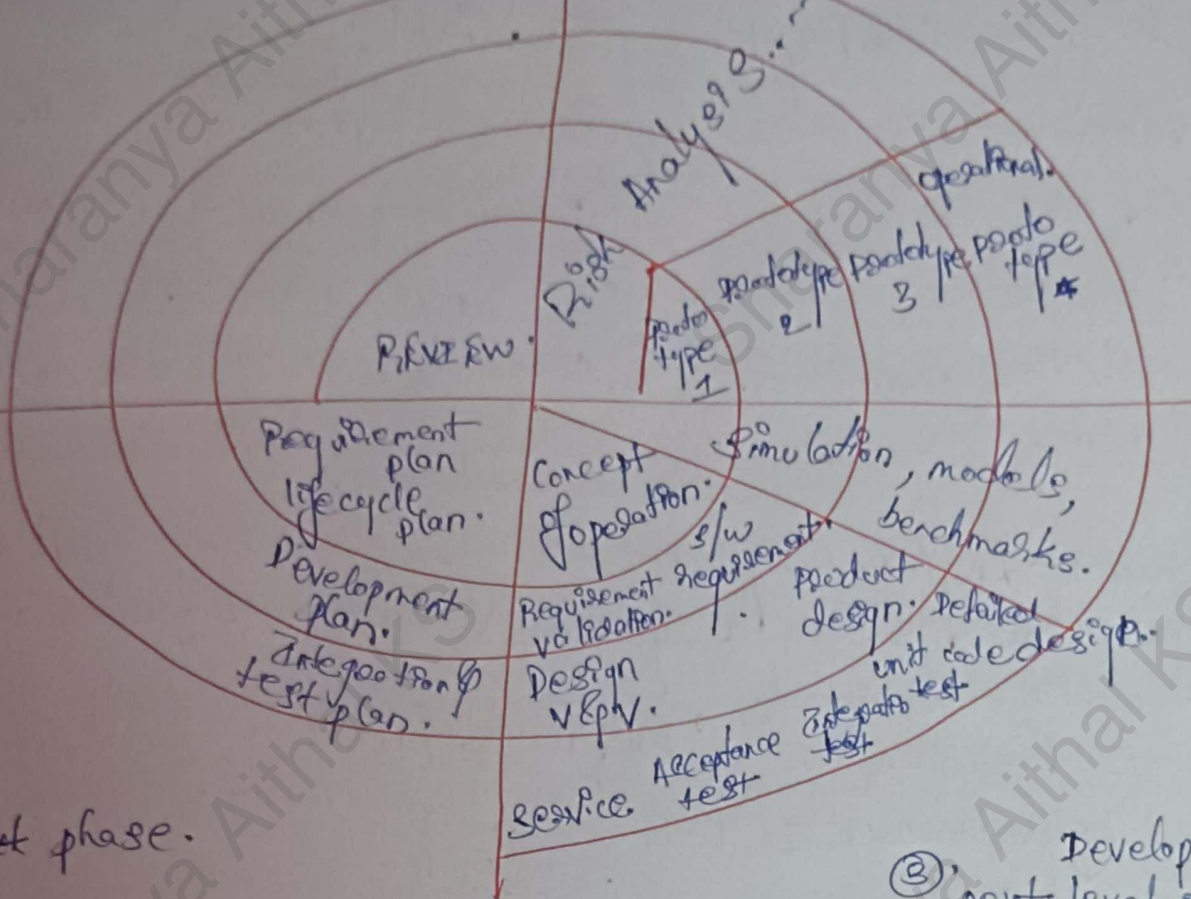


Determine objectives, alternatives & constraints.

①

Evaluate alternatives

② identify, resolve risks.



④ plan next phase.

③ Develop, verify next level product.

## Spiral model

① Risk driven software development model. that handles risks early & can manage large projects.

② It is costly.

③ overlapping of phases is not possible.

④ large team is required.

## Incremental model

① Incremental model develops the product in smaller, manageable modules with easier changes.

② It is less costly.

③ overlapping of phases is possible.

④ large team not required.



## Waterfall model.

- ① follows planned driven approach.
- ② Suitable for safety critical systems
- ③ user involvement is less
- ④ maintenance cost is high
- ⑤ user get product at last stage.
- ⑥ process activities organised in sequence.

## Incremental model.

- ① follows either plan driven or agile
- ② Suitable for application systems.
- ③ user involvement is more
- ④ maintenance cost is low.
- ⑤ user get product at initial stages
- ⑥ process activities are interleaved

**Requirement Engineering:** process of establishing services that the customer requires from a system & constraints under which it operates & is developed.

Types of Requirement: (Ex: MHC-PMS).

User Requirement & System Requirement.

Client managers  
System end users  
Client engineers  
Contractor/manager  
System architects.

System end users  
Client engineers  
System architects  
Software developers.

### Functional

&

### Non functional requirements.

User able to search appointment list.  
List of patient who are expected to attend appointment.  
Staff using system uniquely identified.

only authorize ppl access system  
Product requirement  
Organizational requirements  
External requirements.

## Scenarios.

- ① Initial assumptions
- ② Normal.
- ③ What can go wrong.
- ④ Other activities.
- ⑤ System state of completion.

